skilled artisans contributed, precise instruments & exp method

some noteworthy women in Sci:
(Cavendish, Anne Conway, Mary Astell)
debated Descartes mind-body dualism

Descartes talked w Eliz of Bohemia
(thought she was smart)

16.3

Sci Rev
- questioned Christ
- debates
- questioned old religious certainty
- more spread of ideas & goods
- ↑ ways of life & of thought
- rationalism
- Enlightenment → & Sci Rev to society

early enlightenment
- loosely tied
- 1690-1789 activity in DSO
- origins: opposition to absolutism, religious (vs. ?), Sci Rev
- Dutch Rep key: many Huguenots fled from France after Louis XIV, published tracts challenging auth, were political w/c religious

→ Pierre Bayle (1647-1706) - Huguenot in Dutch, examined religious persecution in "Historical and Critical Dict" (69)

shared human beliefs can be misguided
everything doubted → skepticism ~~[illegible]~~ his Dict
~~[illegible]~~ most popular book in 18cen France priv library

Baruch Spinoza (1632-1677) - transition SciRev → Enlght
- apply rat philo to society
- Descartes rationalism, reject dualism
- espouse monism → mind & body same thing
  God nature same thing
- deterministic uni
- good & evil → relative values
- human actions shaped by external, no ~~free~~ free will
- excommunicated by Amsterdam Jews (against relig)
~~attacked by son~~
- model of virtue & intellectual autonomy

Gottfried Wilhelm von Leibniz (1646-1716) optimism
- dev calculus - w use for it
- refuted Cartesian duality & monism
- monads - ∞ substances make up matter in Theodicy 1710
- ours must be best possible uni (good god)
  ↳ mocked by Voltaire Candide on Optimism 1759

period of intellectual turmoil
← member of Royal Society
↳ Locke's Essay Concerning Human Understanding (1690) - most important of time
↳ theory about how humans learn & think
1st principles given by god, tabula rasa a blank tablet

Sensationalism - ideas & thoughts from
sensory impressions

great inspirators of Enlightenment
- Newtons laws
- Lockes [illegible]
- political theory - Locke's "2 Treatises of Gov" 1690
Parliament > crown

Influence of Philosophes

big spread of answers to similar issues
due to philosophes
↳ group of Fr intellectuals who
said they were bringing the light of
reason to ignorant fellow humans

in 1750, Fr center of E civz...
- lang of edu = Fr
- wealthiest & most pop
- ↑ unpopularity for Fr monarchy calls for reform
among edu elite
- philosophes wanted to reach many elites
who were joined together

↳ Repub of Letters (from Renais)
more int 18c in ideas

imagined transnational realm
where [illegible] discussed

to appeal & avoid censors
wrote plays, novels, ... encyclo
filled w double meanings & satires

baron de Montesquieu (one of great philosophes)
- pioneered in Persian Letters 1721
↳ letters written from POV of Persians judging Euro customs

didn't like absolutism Louis XIV inspired by ~~[illegible]~~ physical sciences
↳ apply critical method to gov
spirit of Laws 1748

gov't shaped by geo & hist
3 types: monarchs, repub, despotisms
likes UK parliament not democrat
↳ didn't like uned poor
separation of powers didn't question Fr monarchy
didn't want tyranny
↳ impacted constitutions US 1787 & Fr 1791

other philosophes
Francois Marie Arouet - pen name Voltaire
- the witty ~~[illegible]~~ [illegible] - mixed w royalty, died rich millions
early career turbulent arrested 2x for insulting nobles

shared Montesquieu's like of UK gov (women)
met madame Chatelet ← (companion
↳ studied sci, 1st translation of Newtons Principia-FC
excluded from Royal Society (still used today)

Voltaire ~~cont~~

praised UK whiggism Cirey

political reformer

best gov = good monarch

↳ humans "rarely worthy to self gov"

didn't believe in equality

only realizable → ppl depend on

laws to protect weak from strong

deism radical

↳ G clockmaker

hated religious tolerance → fanaticism & cruelty

strength of philosophes from dedication and org

Diderot & d'Alembert's Encyclo

72,000 articles

everything discussed

after 1770 ppl attack philo faith in progress

↳ Rousseau - born poor, Parlement by skill

rationalism bad, warm & cold feeling balanced

childish goodness

protected from cruel civ

↳ influence Romantic movement

The Social Contract 1762

Enlightenment across Europe

Ppl all over Euro conversed w letters
Voltaire wrote ~8k letters

Republic of letters - international
group of scholars from Americas to
Russia & Euro to Africa & Asia

Some groups focused: sci or religious
↳ reconcile
reason w/ faith

Scottish E
edinburgh →
common sense & sci reason
after Act of Union w UK 1707
- freed from political crisis
- 1st public edu system in Euro

main guy
David Hume (1711-1776) - public good & religious skepticism
mind - processes senses.
reason must be verified with sensory exp
(math or exp)

includes emotions in moral principles
human experience > abstract

influence → Adam Smith (1723-1790)
social interaction → ethics despite natural greed

19C commercial life taught → make civic virtue through competition
fair play / autonomy